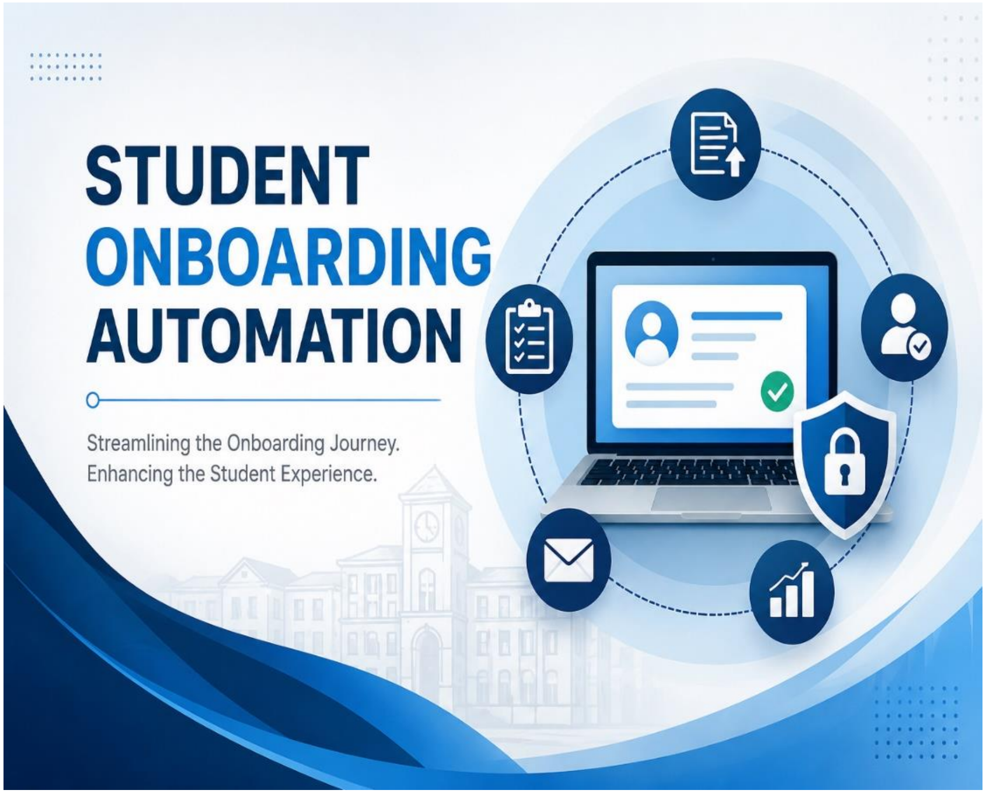

Document Title	Business Requirements Document – Student Onboarding Automation
Organization	The British University in Egypt
Prepared by	Merna Hussein

• Executive Summary

Currently, the university manages the student onboarding process through a combination of manual procedures, spreadsheets, email communications, and disparate systems. While these systems have previously supported onboarding activities, they are becoming progressively ineffective and difficult to manage as student enrollment volumes increase.

The current procedure requires extensive coordination among several departments, including admission, finance, and academic services. The reliance on manual intervention frequently results in processing delays, communication inconsistencies, duplication of efforts, and lack of real-time visibility into the status of student onboarding activities across departments.

To overcome these issues, the university recommends implementing a Student Onboarding Automation System. The proposed solution will provide a digital platform that automates and simplifies onboarding activities from admission acceptance through student activation. It aims to increase operational efficiency, improve student experience and reduce administrative workload across all participating departments.

• Problem Statement:

The current student onboarding process relies heavily on manual effort, disconnected communication methods, and separate administrative systems that do not effectively share information. As a result, university employees struggle to coordinate operations across departments while newly admitted students frequently face delays in completing onboarding requirements.

Key issues identified within the existing process include the manual submission, review, and validation of student documentation, heavy reliance on email-based communication, repetitive data entry and absence of centralized onboarding tracking system. These challenges result in duplication efforts, increasing process time, communication inefficiencies, increasing the risk of errors and data inconsistencies.

Project Overview:

The Student Onboarding Automation Project aims to develop a centralized digital platform that supports the onboarding journey for newly admitted students. The proposed solution will help students with a self-service portal as it will allow them to complete onboarding activities, including account creation, document submission, status tracking, payment, and communication with related university departments. It will also provide automated notifications and real-time updates throughout the onboarding process.

From an administrative perspective, the solution will enable the university departments to manage onboarding activities. Admission, finance, and academic services will be able to review submissions, process approvals, track progress and create student record through a unified platform.

Business Objectives

The primary objective is to establish efficient onboarding process by implementing digital platform. The project aims to achieve the following business objectives:

1- Improve operational efficiency.
2- Reduce manual data entry and human errors.
3- Reduce onboarding cycle time.
4- Enhance student experience.
5- Enhance communication and stakeholder engagement.
6- Increase process visibility and transparency.
7- Improve data accuracy and consistency.
8- Strengthen cross-departmental collaboration.
9- Enhance reporting and decision making capabilities.

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- **Scope**

In scope:

- Digital student onboarding platform:
  - Provide a centralized portal that enables newly admitted students to complete onboarding activities, submit required information, upload documents, and monitor onboarding progress.
- Automated onboarding processes:
  - Automate onboarding workflows, approvals, validations, and related onboarding activities across relevant university departments.
- Electronic document management:
  - Enable the digital submission, review and secure storage of student onboarding documents.
- Automated communication and notifications:
  - Provide automated notifications and status updates to ease communication between students and university staff.
- Integration with existing university systems:
  - Integrate the onboarding platform with existing ERP, student information system, finance, and other relevant university systems to facilitate seamless data exchange and reduce manual data entry.
- User access management and monitoring dashboards:
  - Allow students, staff, and managers to access the system based on their roles and provide dashboards to view relevant information and track progress.
- Management reporting and business insights:
  - Generate operational reports, onboarding statics and management dashboards to support monitoring and decision-making.

Out of scope:

- Core ERP replacement:
  - The project will integrate with existing ERP system but will not replace or modify core ERP functionalities.
- Learning management system:
  - Courses delivery, e-learning, assessments and examinations not included within the scope of the project.
- Alumni management system:
  - Alumni records and post-graduate services outside of the scope.
- Course Registration and scheduling:
  - Course enrollment, timetable management and academic scheduling processes are not included.
- Graduation and certification processes:
  - Graduation requirements and certificate management are not included.

- ## Stakeholders

Stakeholder	Description	Role
Students	Newly admitted students who use the onboarding system (end users).	Submit required information and documents, track onboarding status.
Admission Department	Admission and registration team.	Review admission records and validate submitted documents for admitted students.
University Management	University leaders.	Provide guidance, approve key decisions, and ensure the project supports university objectives.
Finance Department	Department responsible for handling student financial transactions and payment validation.	Check tuition payments, confirm payment and update payment status during onboarding process.
IT Department	Team responsible for supporting and maintaining university systems.	Manage system implementation, integration, security, maintenance and technical support.
Software Development Team	Team responsible for delivering the project and coordinating implementation activities.	Manage project requirements, design, development, testing, training, deployment, and overall project delivery.

- ## Stakeholder Expectation:

The stakeholders expect the proposed solution to:

- Faster onboarding process.
- Reduce delays throughout the onboarding lifecycle.
- Accurate data handling.
- Easy to use interface.
- Follow the university policies.

- Improve and ease collaboration and communication between departments.

As-Is Process

The current student onboarding process is manual and involves multiple departments operating through separate systems, emails, and spreadsheets.

The current onboarding process flow:

1- Student submit required documents by email or visiting the university.
2- Admission department reviews and validates the submitted documents.
3- If there are any issues the admission department contacts the student via email and request the missing documents.
4- The student resubmits the missing documents.
5- The finance department manually verifies fee payment.
6- The academic service manually creates and updates the student record in the system.
7- Departments track onboarding process using emails, spreadsheets, and separate systems.
8- The student follows up with different departments through emails to track the status updates.
9- Once the system meets the requirements, the onboarding process is finalized and the student is officially enrolled.

Current challenges:

- Heavy reliance on manual communication and administrative activities.
- Repeated data entry across records and multiple systems.
- Time-consuming processing and approval procedures.
- Limited visibility of onboarding progress and status.

- Higher risk of human errors and data inconsistencies.
- Possibility of missing documents or incomplete submissions.
- Lack of communication and coordination among departments.
- Limited capabilities for reporting and performance monitoring.

Process gaps:

The current onboarding process reveals several operational gaps:

- Absence of centralized onboarding system.
- Limited system integration.
- Reliance on manual routes of communication workflow.
- Lack of standardized workflow management.
- Weak reporting capabilities.

To-Be Process:

The proposed solution will provide a centralized and automated onboarding experience for students and university staff.

The future onboarding process flow:

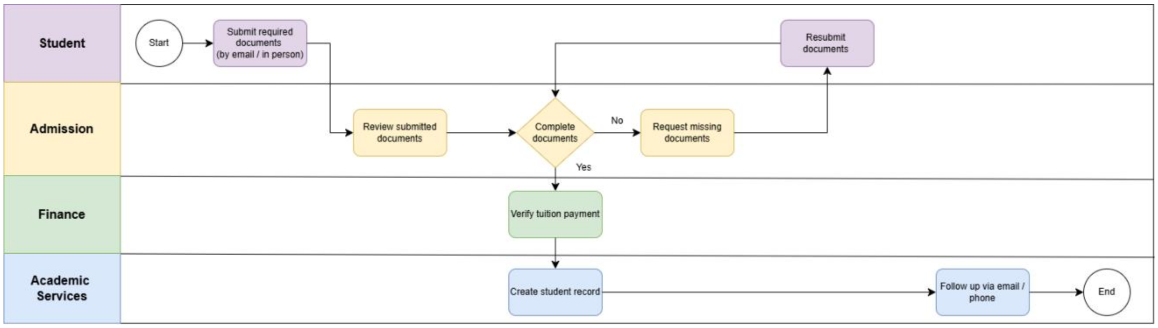
1- The student creates account and logs in the student onboarding portal.
2- The student completes the required information and uploads the required documents through the portal.
3- The system validates the submitted information and verifies that all required documents have been submitted.
4- If any information or documents are missing, the system notifies the student.
5- The admission department review and approves the submitted documents.
6- The finance department verifies the fee payment and update the payment status.
7- The academic service creates or activate the student record and completes the enrollment process.

8- Approved student information is automatically synchronized with ERP and related university systems.
9- Enrollment, financial, administrative records are updated automatically across integrated platforms.
10- The system updates the onboarding status and provides real-time visibility to relevant stakeholders.
11- The student receives automated notification and status updates throughout the process.
12- The student can monitor the onboarding progress through the portal dashboard.
13- Once all requirements are completed, the onboarding process is finalized and student is officially enrolled.

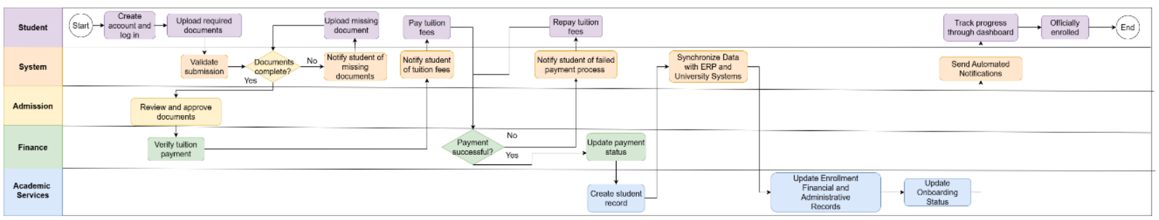
- **Expected Benefits:**
    - Reduced onboarding processing time and administrative effort.
    - Improved operational efficiency through workflow automation.
    - Enhanced student experience through self-services capabilities and real-time updates.
    - Increased visibility and transparency across the onboarding process for all stakeholders.
    - Improved data accuracy and consistency across departments.
    - Strengthened collaboration and information sharing between university departments.
    - Enhanced reporting, monitoring, and decision-making capabilities.

- **BPMN Diagrams:**

  - As-In-BPMN:



  - To-Be-BPMN:



  - To-Be process flow:

1. Student Accesses portal.
2. Student creates account.
3. Student uploads documents.
4. System validates documents.
5. Admission reviews application.
6. Finance validates payment.
7. Academic services activate record.
8. ERP synchronization occurs.
9. Student receives confirmation.

- ## Business Requirements:

Requirement	Priority
Automate onboarding process	High
Reduce manual work	High
Improve communication	High
Improve data accuracy	High

- ## Functional Requirements:

1. Account management:
    - Student shall be able to create an account using admission credentials.
    - Student shall be able to log in to the onboarding portal.
2. Document management:
    - Student shall be able to upload required documents.
    - The system shall validate uploaded documents.
    - The system shall verify that all mandatory documents have been submitted.
3. Workflow and approval management:
    - Admissions staff shall review submitted documents and provide approval.
    - Finance shall verify payments.
    - Academic service shall activate student profile.
4. Notifications and communication:
    - The system shall notify student when documents are missing.
    - The system shall provide notifications throughout the onboarding process.
5. Status tracking and reporting:

- Students shall be able to track onboarding progress though the portal.
- The system shall generate onboarding reports.

6. System integration and security:
    - The system shall synchronize approved student data with ERP and university systems.
    - The system should maintain secure access to student information.

- **Use cases:**

UC-01	Submit Online application
Actor	Student
Description	Student submits application through the portal
Precondition	Student has an account
Post-condition	Submission is stored with unique ID
UC-02	**Upload required documents**
Actor	Student
Description	Student uploads required documents
Precondition	Onboarding application submitted
Post-condition	Documents are stored and linked to the student record

UC-03	Review submission
Actor	Admission officer
Description	Admission staff review submitted information and documents
Precondition	Documents uploaded successfully
Post-condition	Documents are approved, rejected, or correction requested if there is any missing document

UC-04	System synchronization
Actor	System
Description	Approved student information is synchronized with ERP
Precondition	Application approved
Post-condition	Student record created successfully

- ## User Stories:

1- Student:

- As a student, I want to avoid unnecessary campus visits by completing onboarding online.
- As a student, I want to monitor my onboarding progress to know the next required step.

2- Admission officer:

- As an admission officer, I need all submissions available in one dashboard so review can be completed efficiently.
- As an admission officer, I want to request missing documents electronically to reduce manual follow-ups.

3- Finance officer:

- As a finance officer, I want payment verification integrated into the system so the enrollment confirmation can be processed efficiently.

4- System administrator:

- As a system administrator, I want configurable workflows so that future process changes can be implemented without significant redevelopment.

- ## Functional acceptance criteria:

1- Application submission:
   - Mandatory fields must be completed before submission.
   - Data format validation shall be enforced.
   - A unique ID shall be generated automatically.

2- Document Upload:
   - Only authorized file formats will be accepted.
   - File size restrictions shall be enforced.
   - Documents shall be linked to student record.

3- Application review:
   - Staff shall be able to filter submissions by status.
   - Approval and rejection actions shall be recorded with timestamps.

4- System integration:
   - Approved records shall synchronize automatically.
   - Failed synchronization attempts shall be logged and retired.

- ## Technical Requirements:

1- System design:
   - System shall be developed as a web-based application.
   - Integration shall be API-driven.
   - The system shall support cloud or on premise deployment.

2- Integration:
   - RESTful APIs shall be supported.
   - Secure authentication mechanisms shall be implemented.
   - Error handling and retry capabilities shall be available.

3- Database:
   - A relational database shall be used.

- Data shall be encrypted during storage.
- Backup and disaster recovery mechanisms shall be implemented.

4- User interface:

- The interface shall be responsive across devices.
- Accessibility standards shall be supported.
- English and Arabic languages shall be available.

• Non-Functional Requirements:

Requirement Category	Description
Availability	The system should maintain a minimum availability of 99% during operational hours.
Performance	The system should response in 3 seconds under normal operating conditions.
Capacity	The system should concurrent users during peak admission and enrollment periods without degradation in performance.
Security	The system should provide secure user authentication to ensure authorized access to system functions and data.
Data protection	Data should be encrypted during transmission and storage in accordance with the university's information security standards.
Usability	The system should provide the a user-friendly interface.
Scalability	The system should support the future growth in student enrollment volumes without significant system redesign.

• Reporting and analytics:

- Onboarding status dashboards.
- Processing time reports.
- Student conversion and enrollment metrics.
- Audit and compliance reports.

- Operational performance indicators.

- ## Data objects and field definitions:

  - ### Student onboarding record:

Field	Type	Mandatory
Student ID	UUID	Yes
First name	Text	Yes
Last name	Text	Yes
Email address	Text	Yes
Program	Text	Yes
Intake	Text	Yes
status	Enumeration	Yes

  - ### Document record:

Field	Type	Mandatory
Document ID	UUID	Yes
Student ID	UUID	Yes
Document type	Enumeration	Yes
File location	Text	Yes
Upload date	Date/Time	Yes

- ## RACI matrix:

Activity	Admission	IT	Finance
Define requirements	Responsible	Consulted	Consulted
System development	Consulted	Responsible	Informed
Submission review	Responsible	Informed	Informed

System integration	Consulted	Responsible	Consulted
Payment validation	Informed	Consulted	Responsible
User acceptance testing	Responsible	Responsible	Consulted

- ## Assumptions:
  - Students have access to the internet and devices to use the student onboarding portal.
  - Student possess valid admission credentials.
  - Existing university systems (ERP, SIS, and Finance) support the required integrations.
  - Student data can be migrated.
  - Required technical infrastructure and system resources are available.
  - Required budget and resources for the project are provided.
- ## Risks:

Risk	Solution
User resistance to change	Training and awareness sessions
Integration issues	Early integration testing
Data migration errors	Conduct migration testing
Incomplete requirements	Regular stakeholder reviews
System downtime	Backup plan
Performance issues	Performance testing and capacity before deployment.

- **Constraints:**
  - Budget limitations.
  - Fixed project timeline aligned with admission cycles.
  - Mandatory integration with existing ERP, SIS, and Finance systems.
  - Compliance with University policies and data privacy regulations.
  - Adherence to existing technology infrastructure and standards.
  - Limited availability of business and technical resources.

- **Error handling and exception management:**

1- Failed payment:
  - Student shall receive immediate notification when payment processing fails.
  - The system shall support configurable retry attempts.
  - Failed transactions shall be recorded for audit purposes.
  - Enrollment confirmation shall not proceed until successful payment.

2- Failed system synchronization:
  - Synchronization failure shall be logged automatically.
  - Retry mechanisms shall be executed according to configured policies.
  - Persistent failures shall be escalated to IT support.

3- Missing or invalid documents:
  - Applications with incomplete documents shall not proceed.
  - Students shall receive requests for corrective action.
  - The application status shall reflect the pending requirements.

• Implementation plan:

1- Phase 1 - Analysis and design (4 Weeks)

- Requirements gathering (week 1)
- Process mapping (week 1-2)
- Solution design (week 2-3)
- UI/UX (week 3-4)

2- Phase 2 – Development and integration (8 Weeks)

- Portal development (week 5-9)
- Workflow implementation (week 6-10)
- Notification services (week 8-10)
- System integrations (week 9-12)

3- Phase 3 – Testing and validation (4 Weeks)

- Functional testing (week 13)
- Integration testing (week 14)
- Performance testing (week 15)
- User acceptance testing (week 16)

4- Phase 4 – Deployment and training (2 Weeks)

- Production deployment (week 17)
- User training (week 17-18)
- Go-live support (week 18)

• Budget consideration:

Estimated costs may include:

- Software development activities.
- Integration services.
- Infrastructure and hosting.
- Licensing requirements.
- Training and support services.

- ## Success Metrics (KPIs):

• KPI	Target
Onboarding time	Reduce by 50%
Manual emails	Reduce by 80%
User satisfaction	Above 90%
Data errors	Reduced by 70%

- ## Compliance and governance:

The solution shall comply with:

- University policies.
- Data privacy regulations.
- Audit and monitoring requirements.
- Information security standards.